

RACHAPPA BIRADAR

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SUMMARY

Computer Science graduate with hands-on experience in backend development, embedded systems, IoT, Linux administration, and cybersecurity labs. Experienced in building Node.js/Express REST APIs, multi-protocol IoT systems, and self-hosted infrastructure using Docker, Proxmox, PostgreSQL, and Cloudflare Tunnel. Teaching Assistant at PES University supporting IoT and ethical-hacking coursework and hands-on security labs.

TECHNICAL SKILLS

Programming: JavaScript, Java, C, C++ ,Bash

Backend: Express.js, REST APIs, WebSocket, Socket.IO, JWT

Databases: PostgreSQL, MongoDB, MySQL, Redis

Embedded / IoT: ESP32, Raspberry Pi, Arduino, MQTT, Zigbee, LoRa, Bluetooth

Networking / Security: Firewalls, RBAC, OS Hardening, Nmap, Wireshark, Burp Suite, Wazuh, Metasploit

Infrastructure: Linux/Ubuntu, Docker, Proxmox, systemd, Tailscale, Cloudflare Tunnel

PROFESSIONAL EXPERIENCE

Teaching Assistant — IoT & Embedded Systems — PES University, ISFCR 2026 – Present

- Assist in teaching an embedded systems course, building lecture and lab material covering ESP32 architecture, memory, GPIO, and communication protocols (UART, SPI, I2C, CAN, Wi-Fi, Bluetooth).
- Authored a professional IoT project proposal with a custom architecture diagram (ESP32-C3 sensor nodes, MQTT/CoAP, Redis, PostgreSQL) used as course reference material.

Teaching Assistant — Ethical Hacking & Exploit Dev. — PES University, ISFCR (M.Tech) 2026 – Present

- Support the “Ethical Hacking and Exploit Development” course for Executive M.Tech students, designing lab exercises and grading ISA/ESA assessments.
- Built and maintain a dedicated TryHackMe room and a live target domain for student reconnaissance exercises covering passive/active recon, Wireshark analysis, and Nmap scanning; conducted hands-on workshops on phishing, Wi-Fi Evil Twin, and call-bombing attacks.

PROJECTS

Secure Multi-Protocol IoT Gateway (Capstone) — Node.js, MongoDB, MQTT, Raspberry Pi, Arduino

- Designed and built an embedded IoT gateway that collects real-time telemetry from Raspberry Pi and Arduino sensor nodes, handling ingestion for multiple concurrent device connections without data loss.
- Integrated four communication protocols — Wi-Fi, Bluetooth, Zigbee, and LoRa — into a single gateway, using MQTT as the core low-bandwidth messaging layer for constrained hardware.

AI-Based Digital Addiction Relapse Prediction System — Node.js, Express.js, Python, Scikit-learn

- Designed backend REST APIs (Node.js, Express.js) to collect, validate, and process real-time usage-log data from client applications.
- Engineered a Python feature pipeline (Usage Velocity, Activity Ratio, App Entropy) and trained a Random Forest model (ROC-AUC 0.80), served via a real-time prediction API for relapse-risk scoring.
- Tuned hyperparameters via cross-validation and evaluated precision/recall alongside ROC-AUC to account for class imbalance between relapse and non-relapse cases.

Real-Time ESP32 Motion Capture System — ESP32, IMU Sensors, MQTT, NTP, Redis, WebSockets

- Built a real-time motion-capture pipeline using multiple IMU sensors, synchronized via NTP and streamed continuously over MQTT to a central server.
- Used Redis for buffering and frame synchronization across sensors, and WebSockets to stream synchronized frames to a browser-based 3D skeleton visualizer for live motion analysis.

Cybersecurity Lab & Self-Hosted Home Lab — Proxmox, Ubuntu Server, Docker, Wazuh, Nmap, Burp Suite

- Built isolated virtualized lab environments on Proxmox for security experimentation, alongside a self-hosted home lab for servers, networking, and web applications/APIs.
- Used Wazuh for security monitoring and log analysis, and Nmap and Burp Suite for reconnaissance and web-security testing in controlled lab environments.

EDUCATION

B.Tech in Computer Science — PES University, Bengaluru

CGPA: 6.7